



Metagenomics sample classification

Bartosz Smolarczyk, Wiktor Grzankowski



Core Algorithm for Metagenomic Classification

Objective

- classify metagenomic samples by generating k-mer profiles from DNA sequences and using them for classification

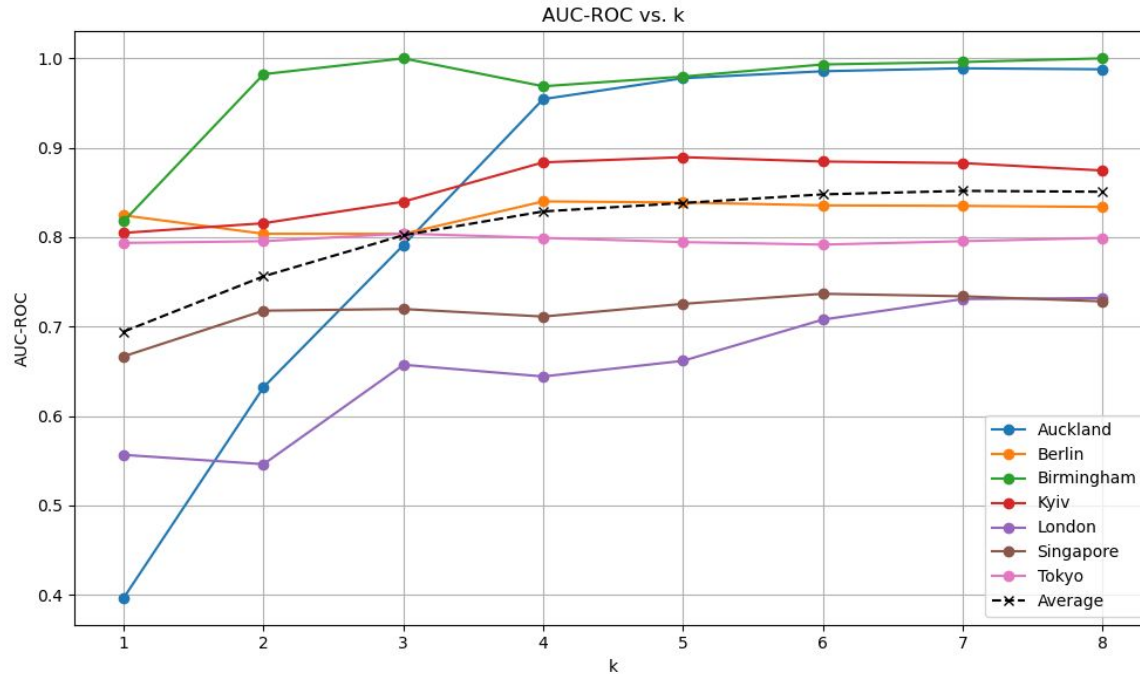
Steps in the algorithm

- Load data - parse training and testing datasets and labels
- Generate k-mer profiles (important - what length?)
 - Extract k-length substrings (k-mers) from DNA sequences
 - Count the occurrence of each k-mer in the training data
 - Downsample high-frequency k-mers to mitigate noise
- Aggregate class profiles
 - Combine k-mer profiles across all samples for each class
- Classify testing samples
 - Generate k-mer profiles for each test sample
 - Compare test profiles with class profiles using cosine similarity
 - Normalize scores to generate likelihoods for classification

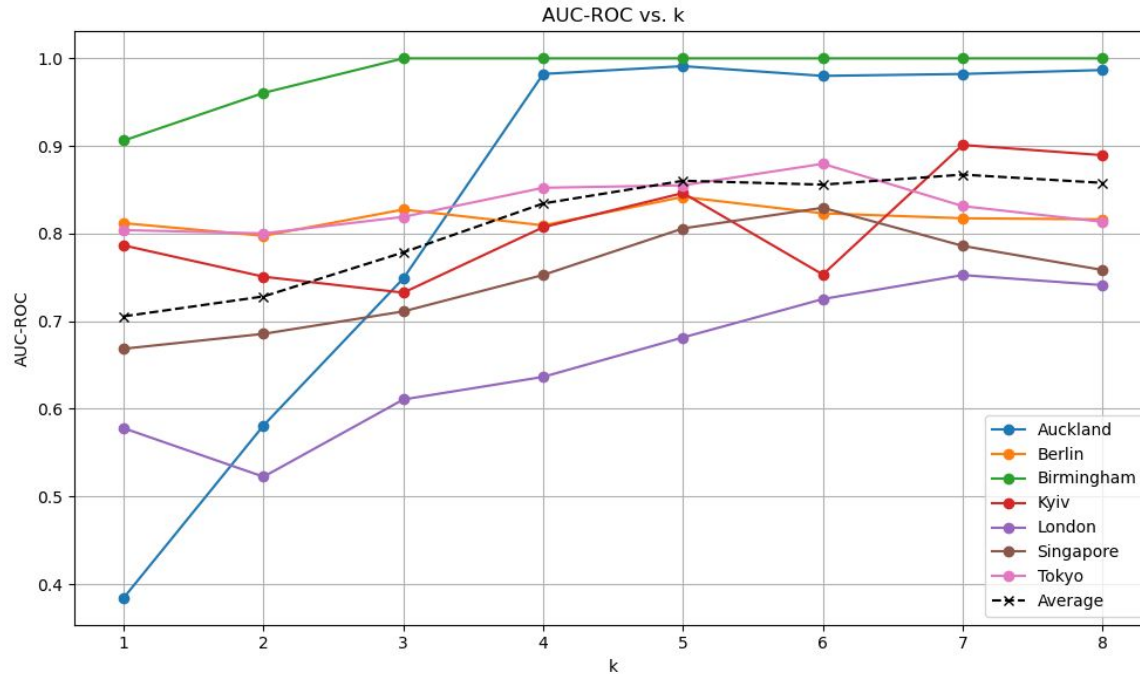
Choosing the right size of k-mers

- For greater values of k the results are more stable and accurate
- For greater values of k the runtime grows exponentially
- “data noise” should be considered, different for each k (we do not like it)
- Best way to find the right value? Try them all...
- Turns out to be $k = 7$

AUC-ROC vs. k (sample)



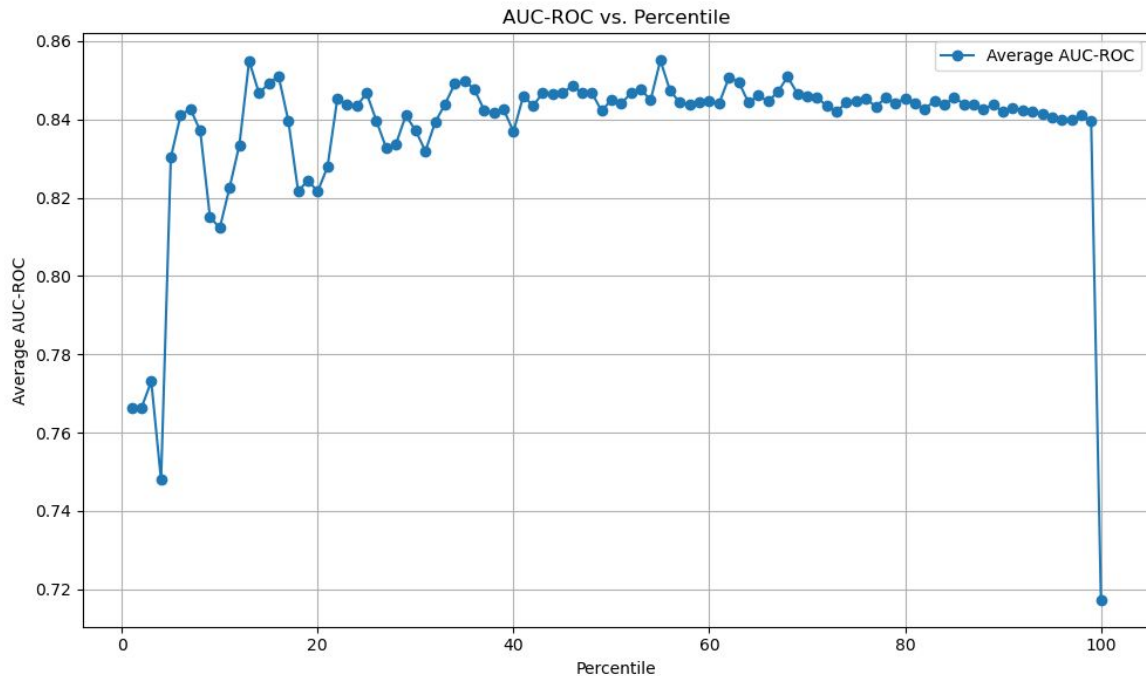
AUC-ROC vs. k (large)



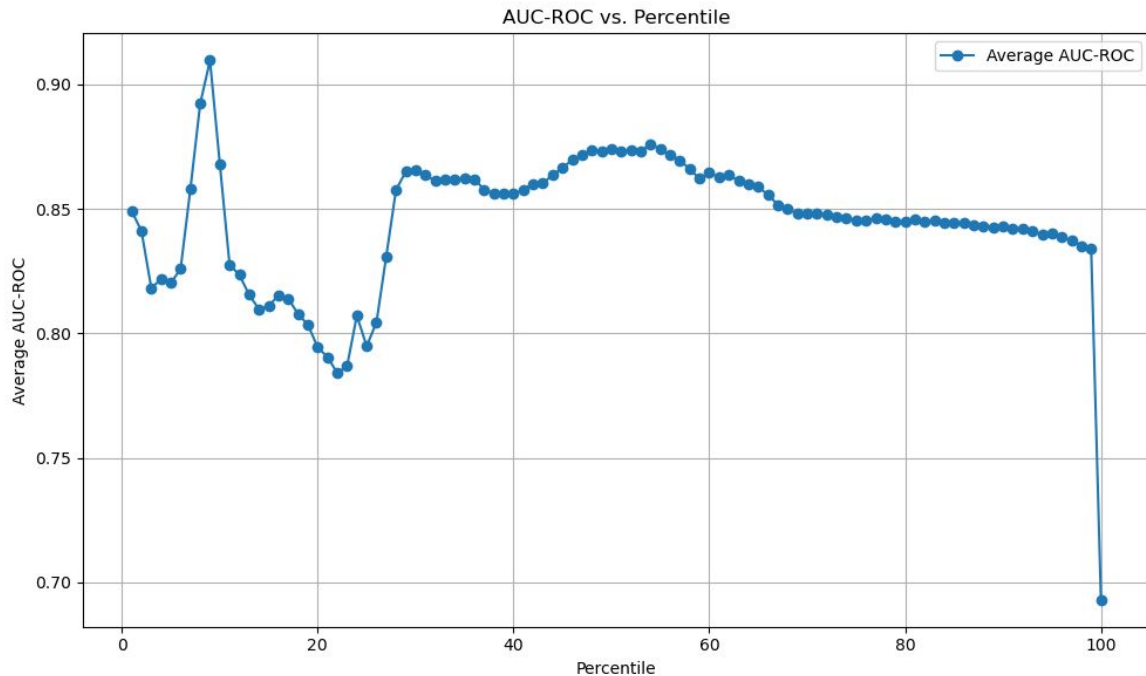
Eliminating data noise

- The occurrence counts for different k-mers vary greatly
- For some important, unique k-mers, there may be just a few occurrences
- For some common, shared k-mers, there may be a lot of occurrences
- How to strengthen the impact of unique k-mers while keeping the significance of common k-mers?
- Downsample the counts by the chosen percentile (again, experimenting with all sorts of values...)
- $p = 65\%$ was considered the safest (notice how the chart oscillates for $p < 40\%$ and drops for $p > 90\%$)

AUC-ROC vs. percentile (sample, $k = 7$)

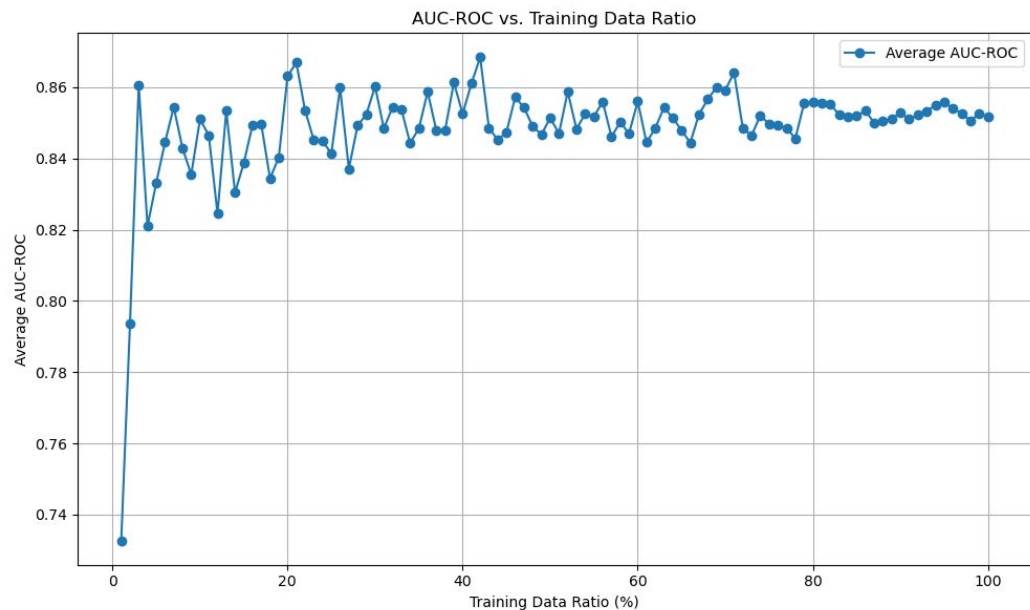


AUC-ROC vs. percentile (large, $k = 7$)



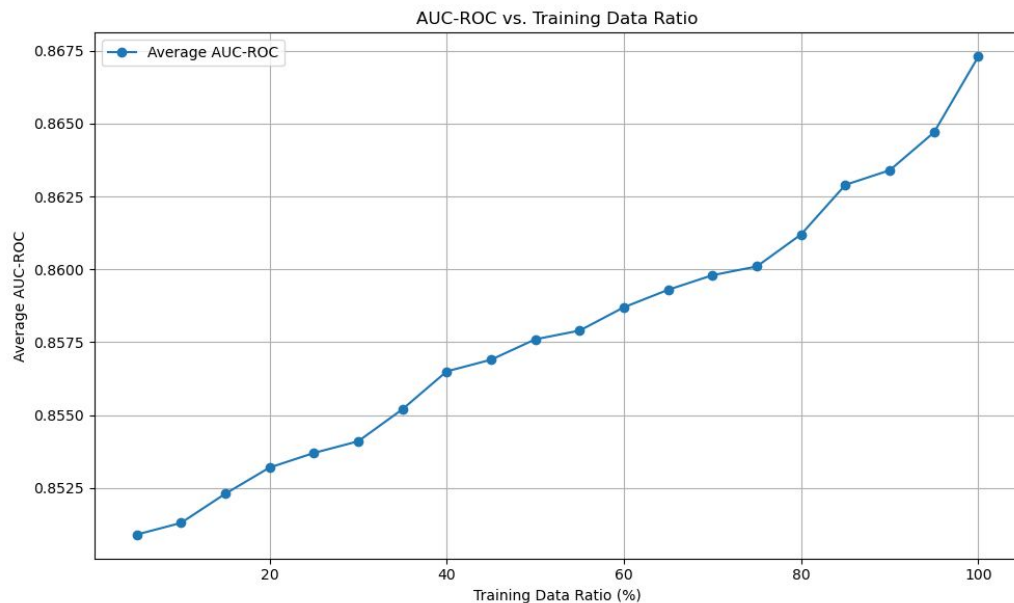
Additional items

downsample training data (sample, $k = 7$, $p = 65\%$)



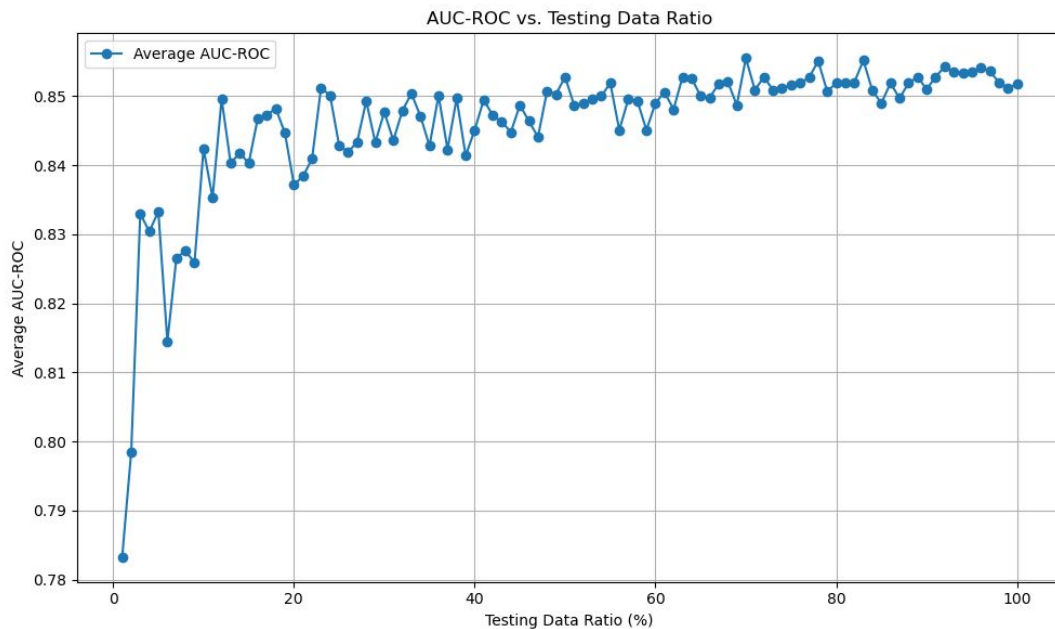
Additional items

downsample training data (large, $k = 7$, $p = 65\%$)



Additional items

downsample testing data (sample, $k = 7$, $p = 65\%$)



Additional items

downsample testing data (large, $k = 7$, $p = 65\%$)

